

Year 8 Science – Booklet 8: Physics

Motion & Pressure — ANSWERS

Note: every page in this booklet covers Pressure only – there is no Motion content in the pages provided, so none is answered below.

Quick Test (p.97)

1. By applying a force over a small area (concentrating the force).
2. By spreading the force over a larger area.
3. The weight of the shopping is concentrated onto the small area of the thin handles, creating a high pressure on your hands.
4. Pascals (Pa).
5. A sharp knife has a much smaller (thinner) blade edge area, so the same force creates a much higher pressure, making it easier to cut through.
6. $P = F/A = 50 \div 2.5 = \mathbf{20\ Pa}$
7. $\text{Area} = 4 \times 2 = 8\ \text{m}^2$. $P = 4000 \div 8 = \mathbf{500\ Pa}$
8. $F = P \times A = 40 \times 2 = \mathbf{80\ N}$
9. At the bottom of the lake (the greatest depth) – pressure increases with depth.

Section A – Multiple Choice (5 marks)

1. c) someone wearing snowshoes
2. c) 16 Pa
3. b) 50 N
4. b) 5 m²
5. b) increases with depth

Section B (13 marks)

1. True or false (7 marks)

- a) Thin handles create high pressure, making it easier to carry – **False** (high pressure makes it more painful, not easier)
- b) Snowshoes spread force to create low pressure, stopping you sinking – **True**
- c) Pressure on a nail is greatest on the head – **False** (it's greatest at the sharp point, the smallest area)
- d) Sharp knives create higher pressures than blunt knives – **True**
- e) Press-ups on palms create the same pressure as on fingertips – **False** (fingertips have a much smaller area, so create much higher pressure for the same weight)
- f) Camels have large feet to grip the sand better – **False** (large feet spread their weight to reduce pressure, stopping them sinking – not about grip)
- g) More likely to fall through a frozen lake standing up than lying down – **True** (standing concentrates your weight on your feet; lying down spreads it over a larger area)

2. Calculate the pressure

- a). 50 N over 25 m² = $50/25 = \mathbf{2\ Pa}$
- b). 80 N over 16 m² = $80/16 = \mathbf{5\ Pa}$
- c). 25 N over 2.5 m² = $25/2.5 = \mathbf{10\ Pa}$

3. Calculate the force

- a). over 10 m² for 3 Pa: $F = 3 \times 10 = \mathbf{30\ N}$
- b). over 4 m² for 5 Pa: $F = 5 \times 4 = \mathbf{20\ N}$
- c). over 16 m² for 1.5 Pa: $F = 1.5 \times 16 = \mathbf{24\ N}$

Section C (12 marks)

1. 600 N crate, faces A, B, C (3 m × 2 m × 1 m)

- a). Face A (2 m²): $P = 600/2 = \mathbf{300\ Pa}$
- b). Face B (6 m²): $P = 600/6 = \mathbf{100\ Pa}$
- c). Face C (3 m²): $P = 600/3 = \mathbf{200\ Pa}$
- d). Face B should be in contact with the lawn – it has the largest area, so it creates the lowest pressure (least likely to damage the grass).

2. Drawing pin diagram

- a). Mark H at the sharp point of the pin (smallest area, force concentrated = high pressure).
- b). Mark L at the flat, broad head of the pin (largest area, force spread out = low pressure).